

Hikmah
HACKERS

Week 3: Asking Questions

input() lets the user fill the box

DARUL ULOOM MICHIGAN · FRIDAYS

TODAY

Our plan for 40 minutes

Time	What we do
0–5 min	Warm-up: show your My Profile Card
5–15 min	The big idea: input() asks a question
15–30 min	Type along: input(), int(), and a Salah counter
30–37 min	Your turn + bug hunt
37–40 min	Mini-project: My Chat Bot

WARM-UP

Last week, we filled every box ourselves.

Last week

Same answer every time the program runs.

```
name = "Mohammad"
```

Today

The person using the program fills the box.

```
name = input("Name? ")
```

THE BIG IDEA

input() is like a waiter taking your order.

Ask

It shows your question on the screen.

```
input("Name? ")
```

Wait

The program stops until you press Enter.

```
Name? 
```

Store

The answer goes into a box.

```
name = ...
```

LET'S CODE · TYPE ALONG

Your first input()

TYPE THIS, THEN PRESS RUN

```
name = input("Your name? ")
print("Salaam,", name)
```

YOU SHOULD SEE

```
Your name? Mohammad
Salaam, Mohammad
```

“Stuck”? It's waiting for you. Click the output area, type your answer, and press Enter.

SMALL DETAIL, BIG DIFFERENCE

The space matters

No space

```
input("Name?")
```

The answer gets squashed:
Name?Mohammad

With a space

```
input("Name? ")
```

Clean and easy to read: Name? Mohammad

WATCH OUT

input() always gives you text

TYPE THIS, THEN PRESS RUN

```
age = input("Age? ")
print(age + 1)
```

YOU SHOULD SEE

```
Age? 13
TypeError: can only
concatenate str
(not "int") to str
```

Even if you type 13, the box holds the text "13". Remember "8" + "1" from last week?

LET'S CODE · TYPE ALONG

The fix: int()

TYPE THIS, THEN PRESS RUN

```
age = int(input("Age? "))
print("Next year:", age + 1)
```

YOU SHOULD SEE

```
Age? 13
Next year: 14
```

int() turns text like "13" into the number 13. Then math works.

TEXT VS NUMBERS

What ends up in the box?

Code	You type	In the box
`x = input("? ")`	13	"13" (text)
`x = int(input("? "))`	13	13 (a number)
`x = int(input("? "))`	thirteen	ValueError: needs digits

LET'S BUILD

Salah counter

TYPE THIS, THEN PRESS RUN

```
days = int(input("Days? "))
prayers = days * 5
print("Prayers:", prayers)
```

YOU SHOULD SEE

```
Days? 7
Prayers: 35
```

* means times. 7 days × 5 prayers a day = 35.

YOUR TURN · 5 MIN

Plan the dates for iftar

```
guests = int(input("Guests? "))
dates = guests * 3
print("Buy", dates, "dates")
```

Finished early?

Ask one more question, like the host's name, and use it in the last line.

BUG HUNT

Four mistakes everyone makes

Broken	What happens	Fix
`input(What is your name?)`	SyntaxError	Add quotes
`print(age + 1), no int()`	TypeError: text + number	Use int(input(...))
`Typing ten into int(input())`	ValueError	Type digits: 10
`input("Name?")`	Name?Mohammad	Add a space

MINI-PROJECT · DUE NEXT FRIDAY

My Chat Bot

Required

- Ask at least 3 questions with input()
- At least one asks for a number, turned into a number with int()
- Do some math with it (like their age next year)
- Print a friendly reply that uses every answer

Bonus: Salah Counter

```
days = int(input("Days? "))
print("Prayers:", days * 5)
```

Also print the total rakat. A day has 17 fard rakat.

Turn it in at hikmah-hackers.github.io/submit · due Friday, Nov 20

NEXT FRIDAY

More math: boxes and leftovers

```
print(17 // 5)   # 3 full boxes
print(17 % 5)    # 2 left over
```

Next week Python helps you share things out fairly.